Lecture 8

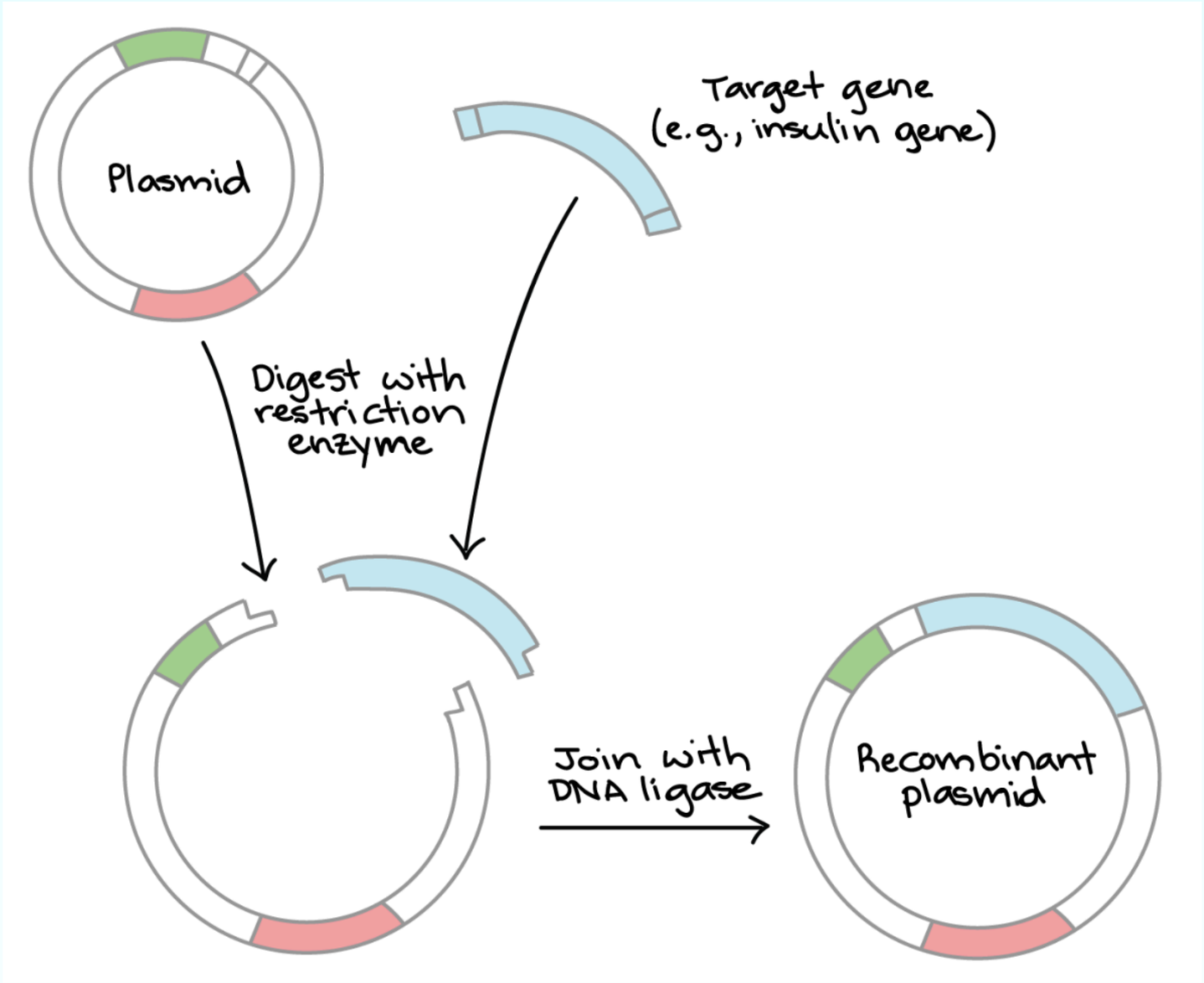
Restriction Enzymes, CRISPR's Cas9, Southern Blot, and Northern Blot

Instructor: Dr. Luu Phuc Loi

Teaching Assistant: Hoang Nhu Quynh

Date: Nov 13, 2026

Recombinant DNA Technology

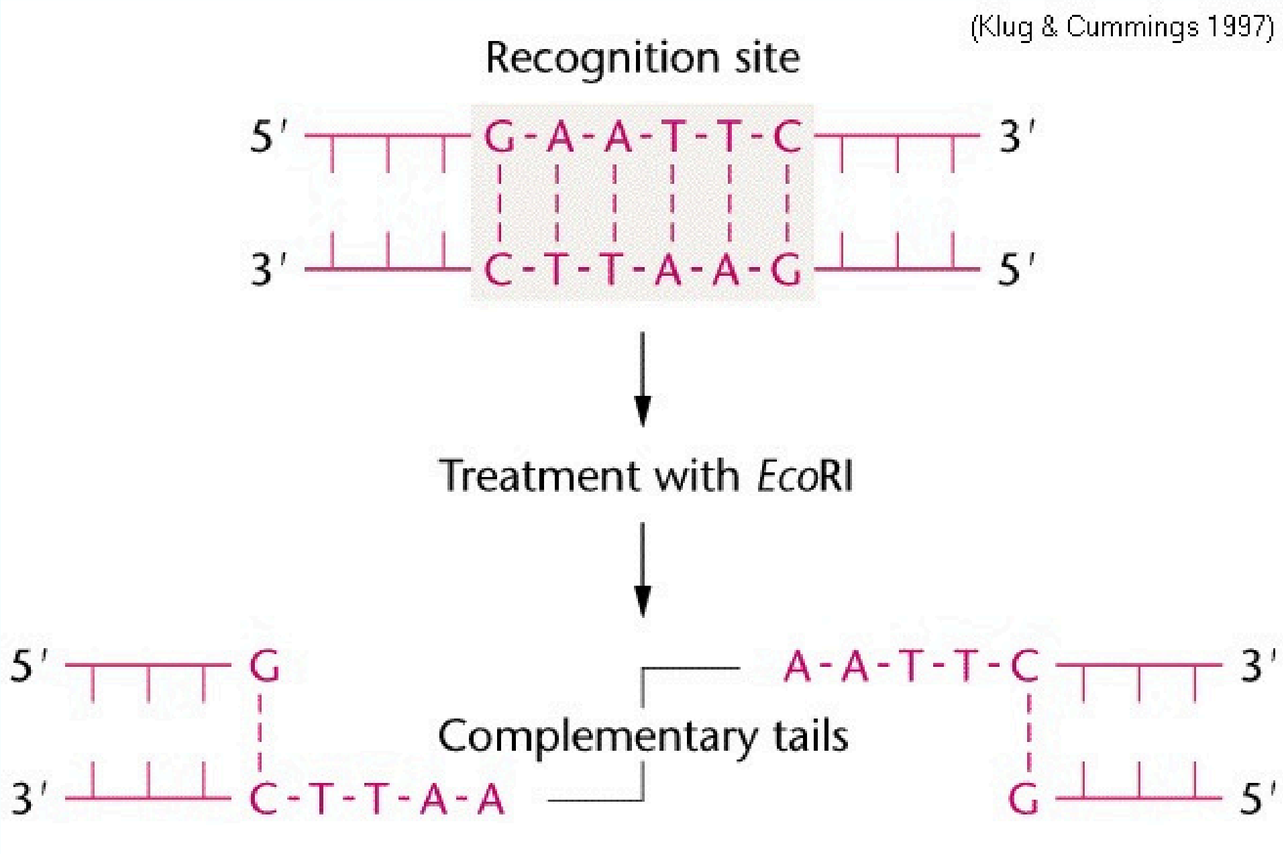


Restriction Enzymes

An enzyme that cuts DNA at a specific sequence

=> Discovered in bacteria

=> cuts at recognition sequence



Types of ends

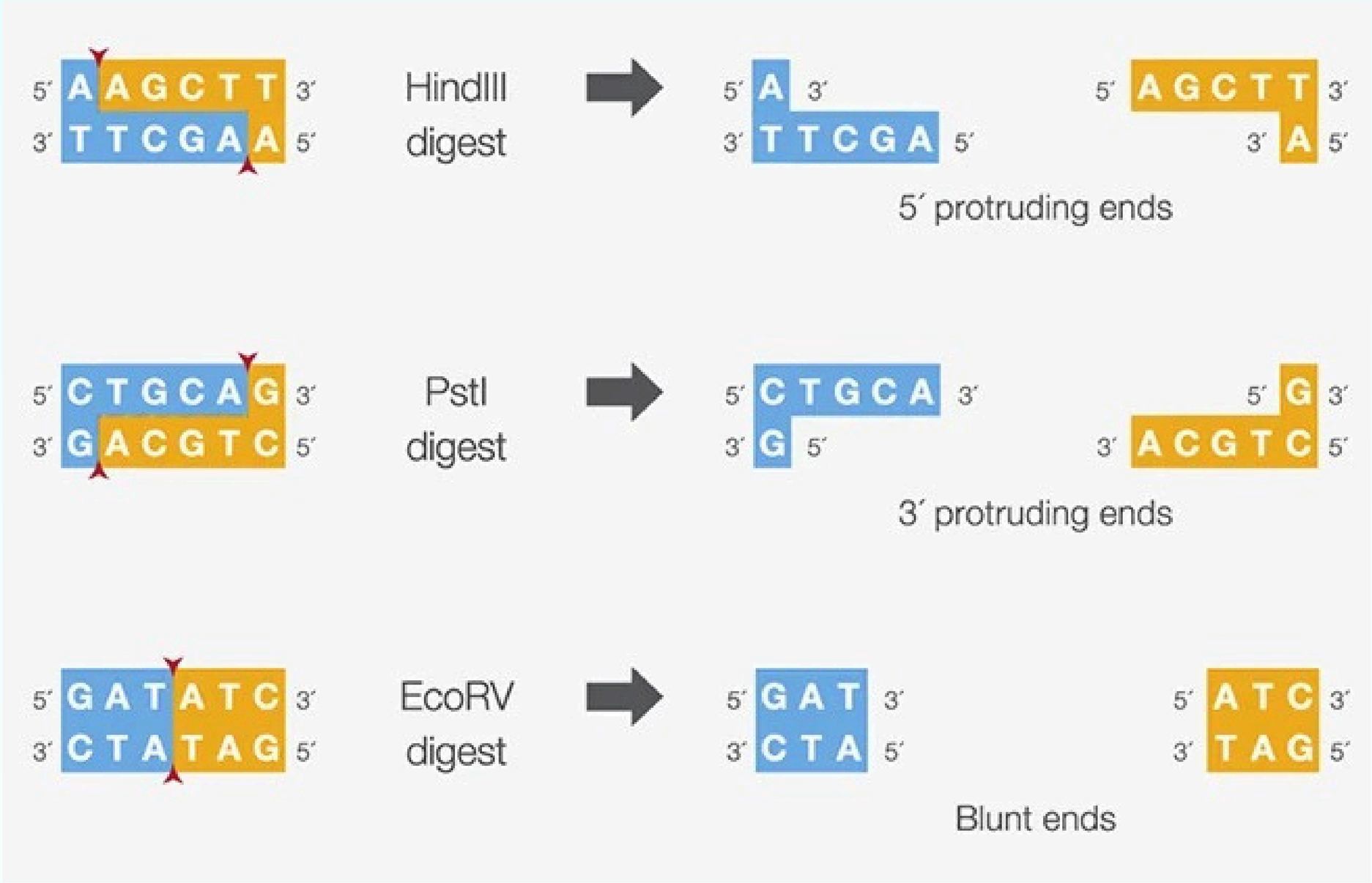
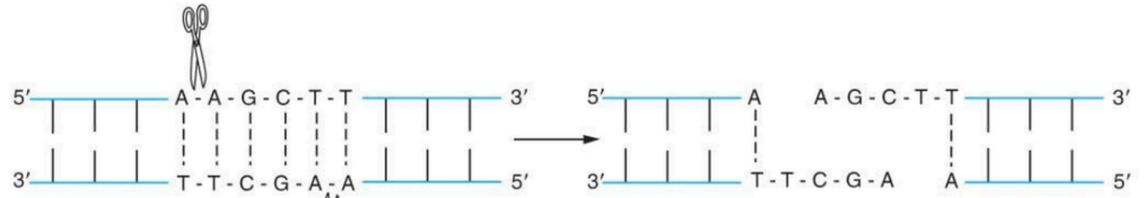
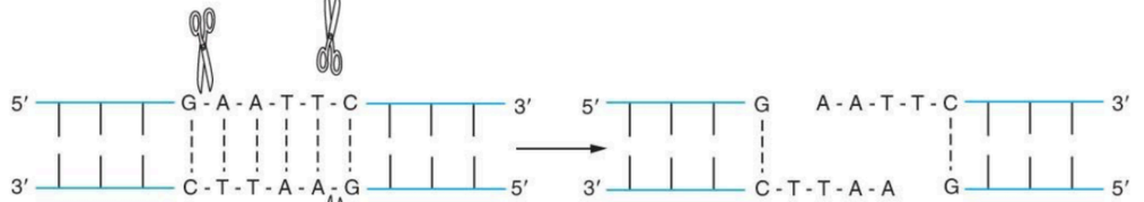
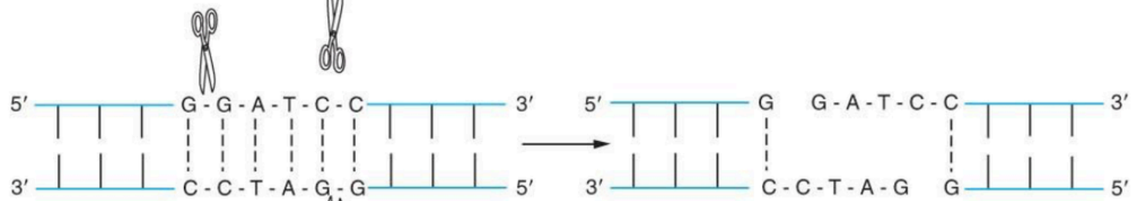
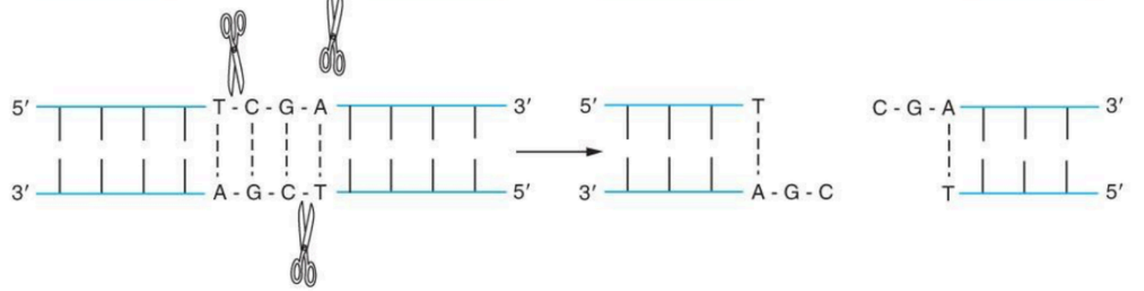


TABLE 3.1 COMMON RESTRICTION ENZYMES

Source Microorganism	Enzyme	Restriction Site
Create Cohesive Ends		
Hemophilus influenzae	*Hin*dIII	
Escherichia coli	*Eco*RI	
Bacillus amyloliquefaciens	*Bam*HI	
Thermus aquaticus	*Taq*I	

Would EcoRI cut the following sequence? Explain.

- **5'CTCGAGTTCGAG3'**
- **3'GAGCTCAAGCTC5'**

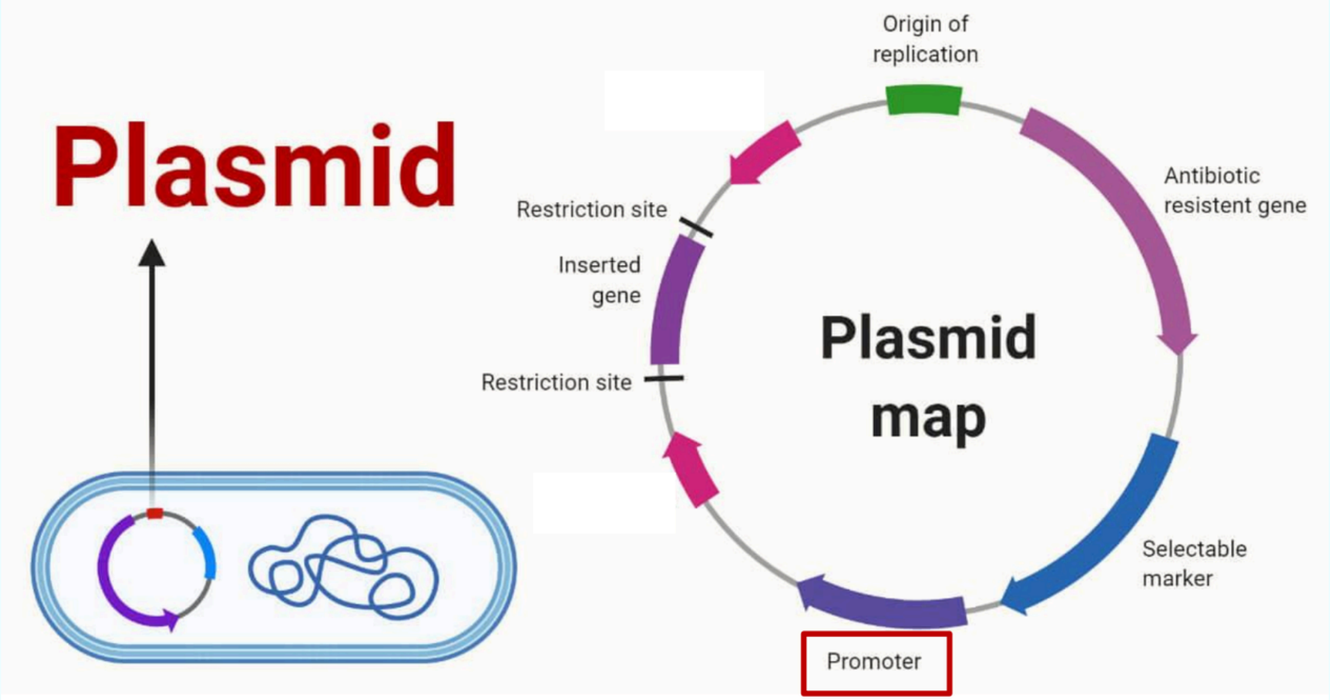
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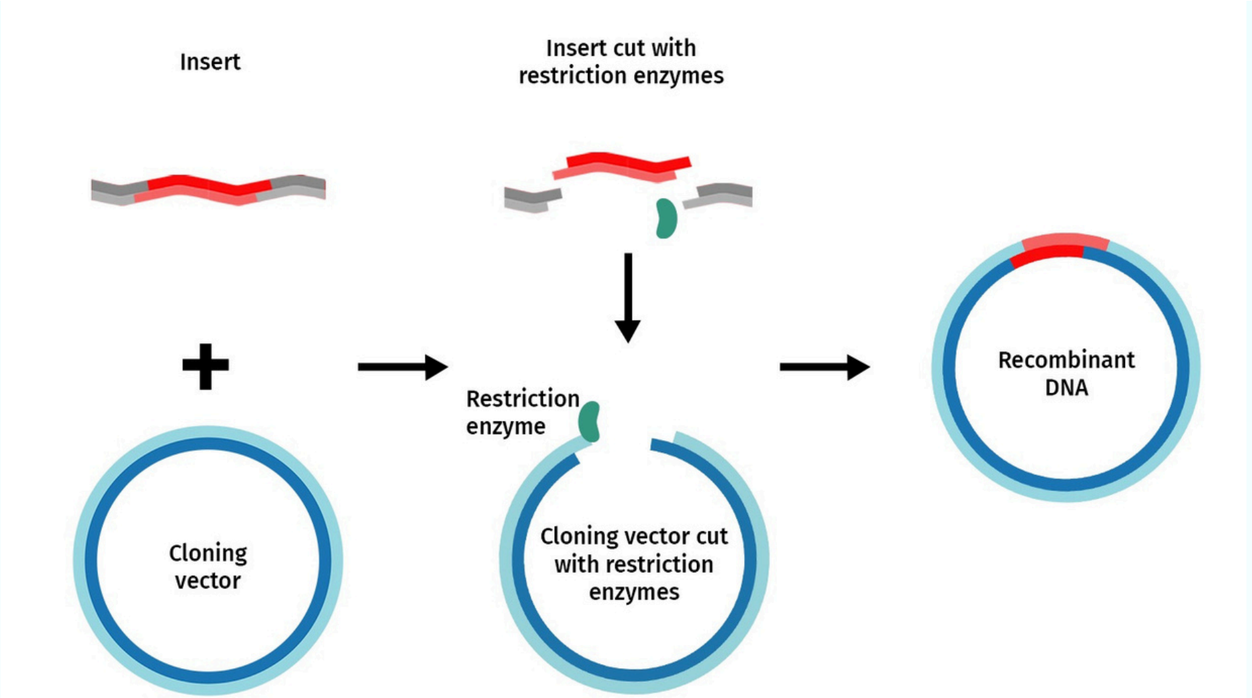
Advantage of enzymes that produce sticky ends

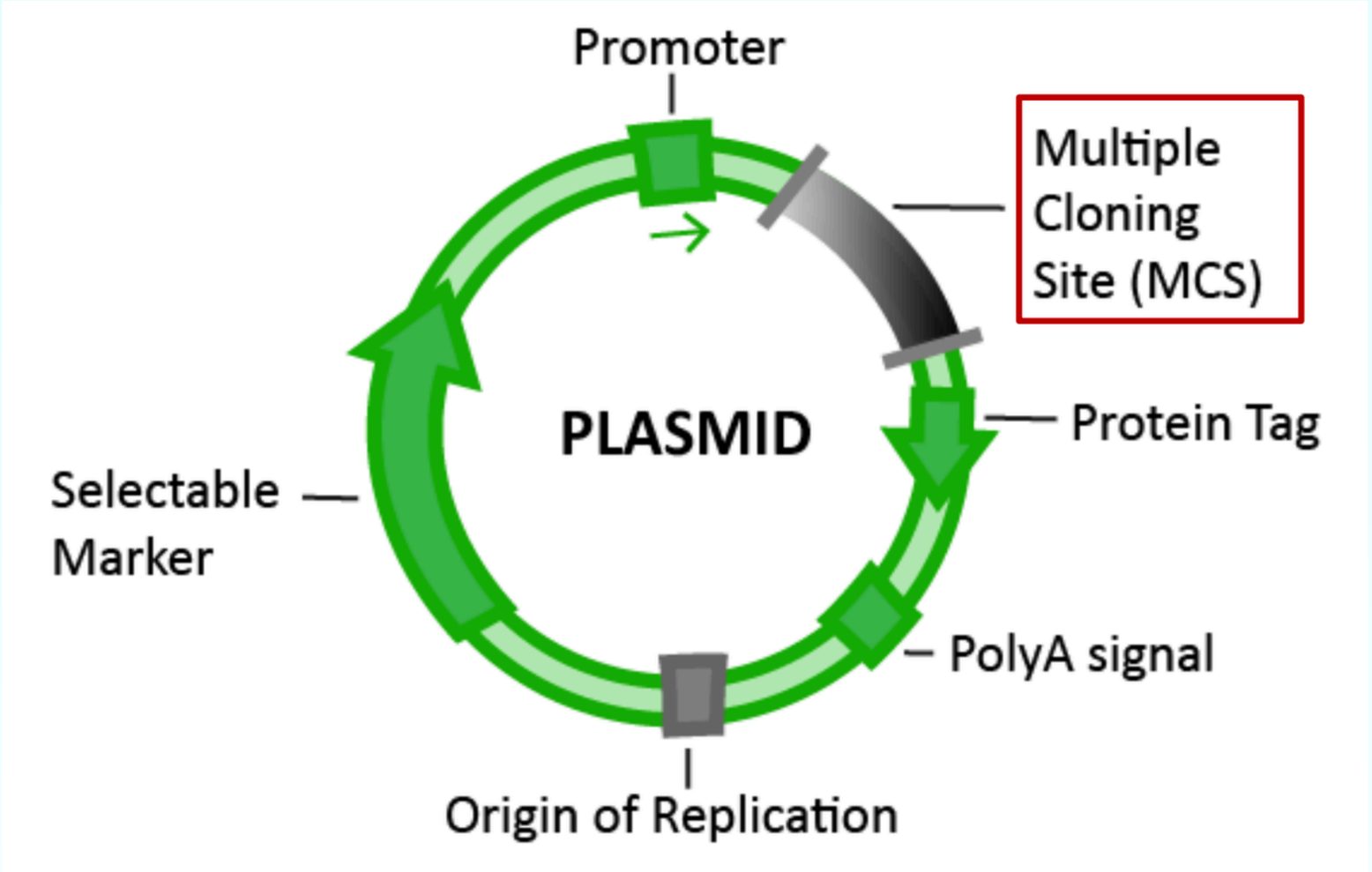
Preferred for cloning because DNA fragments with sticky ends can be easily joined together because they base pair with each other by forming weak hydrogen bonds

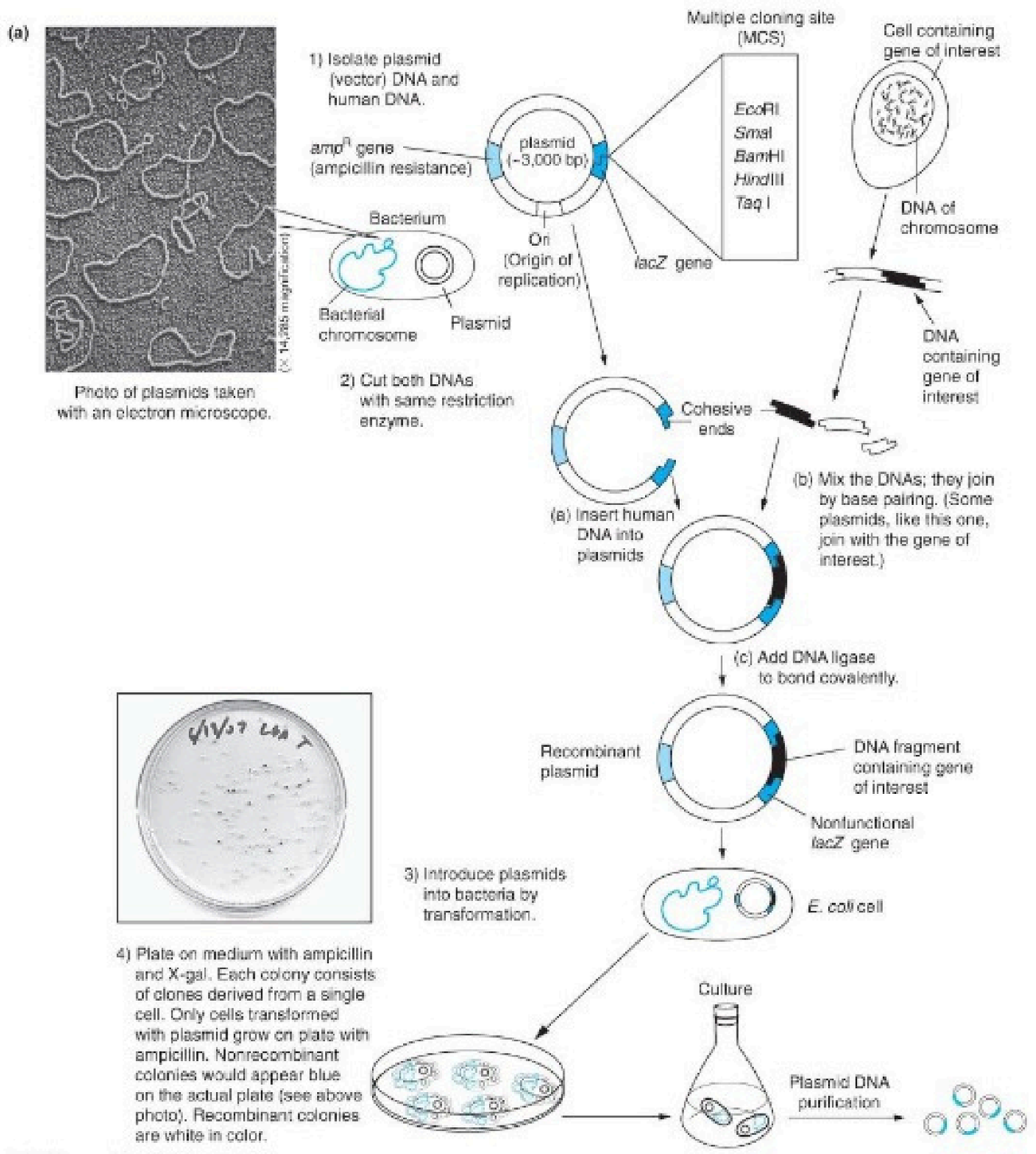
Restriction Enzymes in Cloning

Why don't restriction enzymes digest bacteria DNA ?



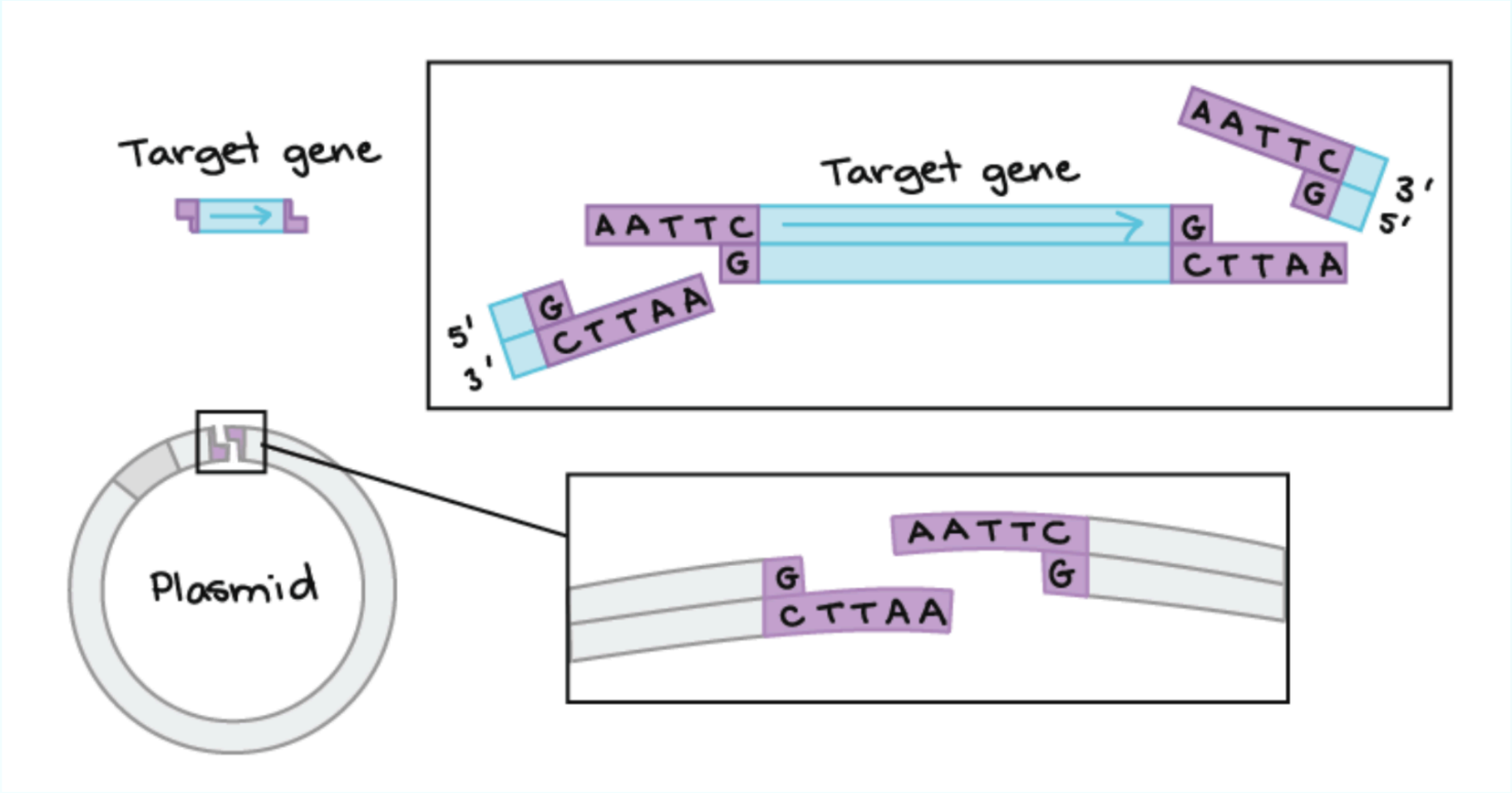



Promoter
Multiple Cloning Site (MCS)
PLASMID
Protein Tag
Selectable Marker
PolyA signal
Origin of Replication



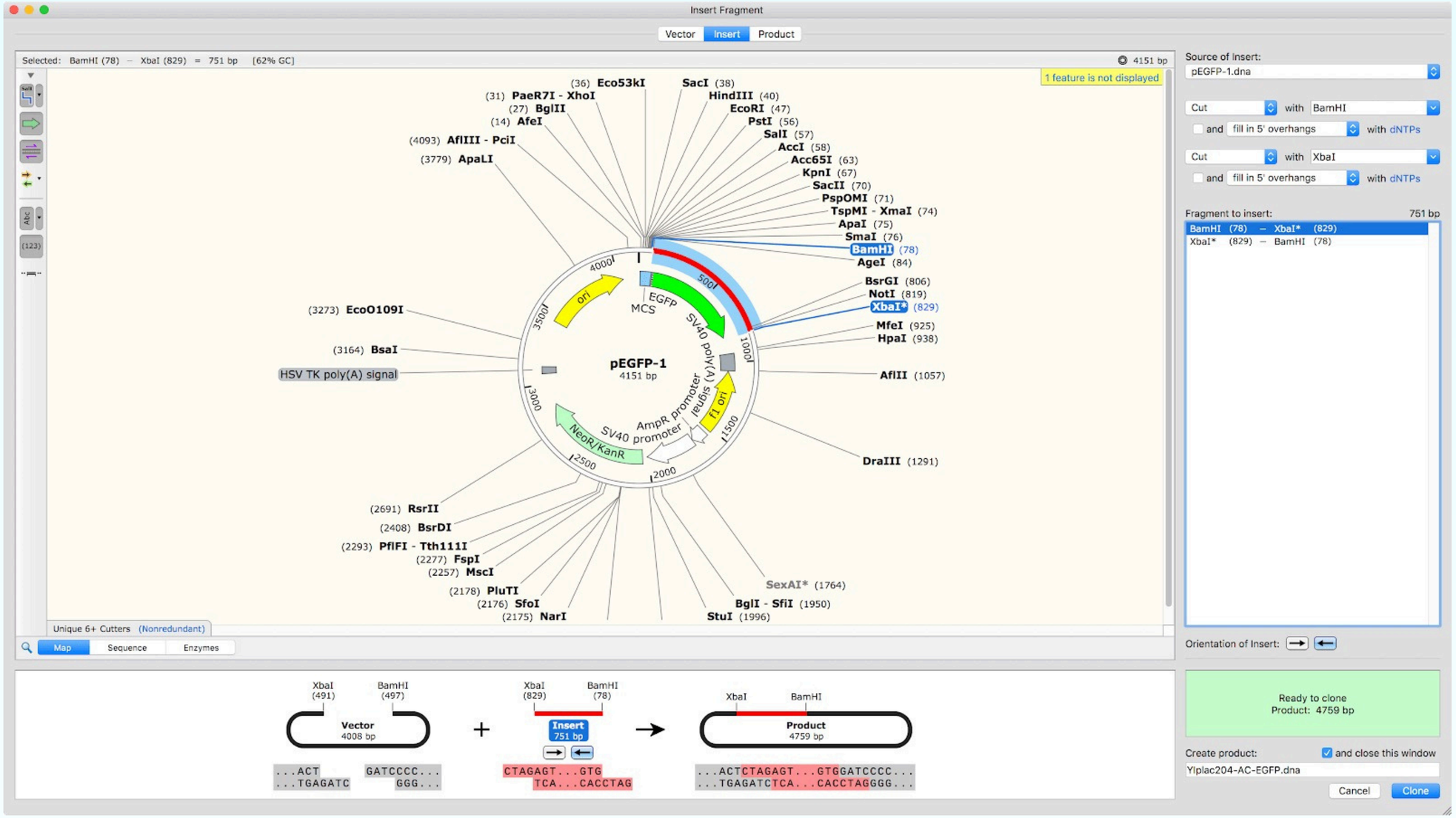
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Criteria to pick-up RE(s)?



Dotmatics

SnapGene



Criteria for selecting restriction enzymes

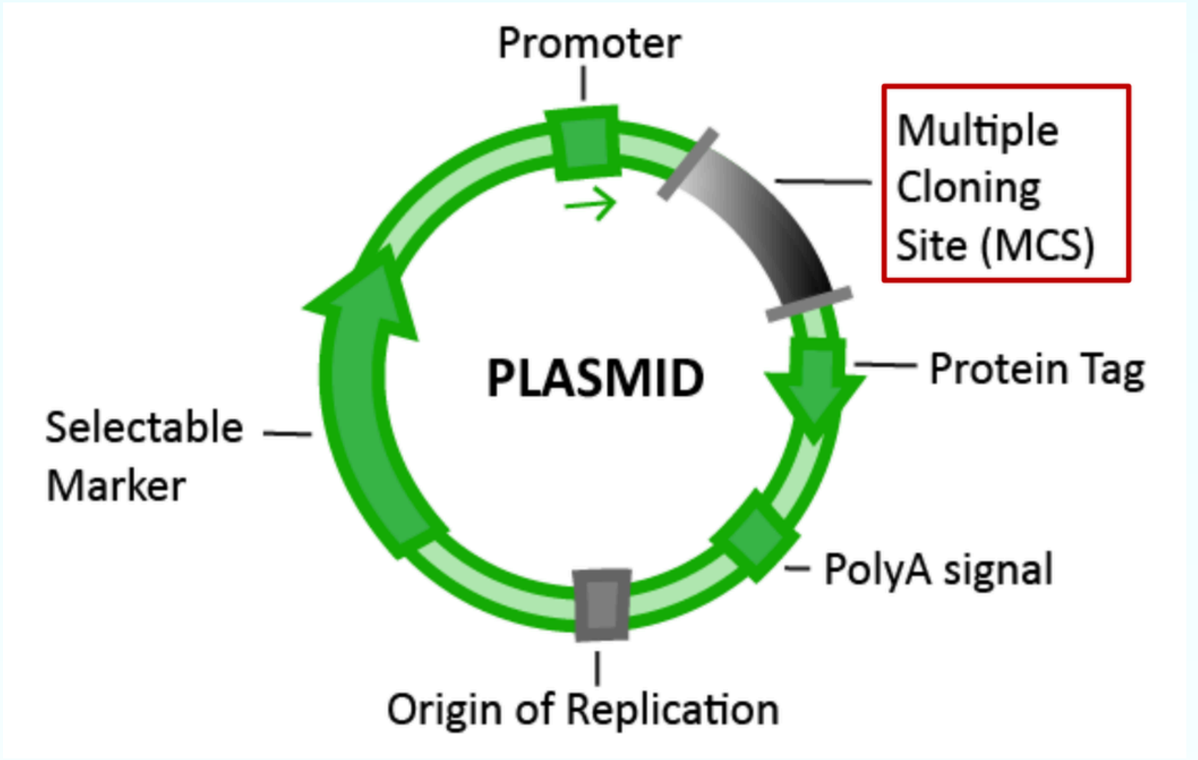
Restriction sites in the insert (GOI)

- The selected enzyme should ideally have no recognition sites inside the GOI
- If an enzyme cuts within your target gene, digestion will fragment your insert and prevent successful cloning of the intact gene.

Criteria for selecting restriction enzymes

Restriction sites in the vector

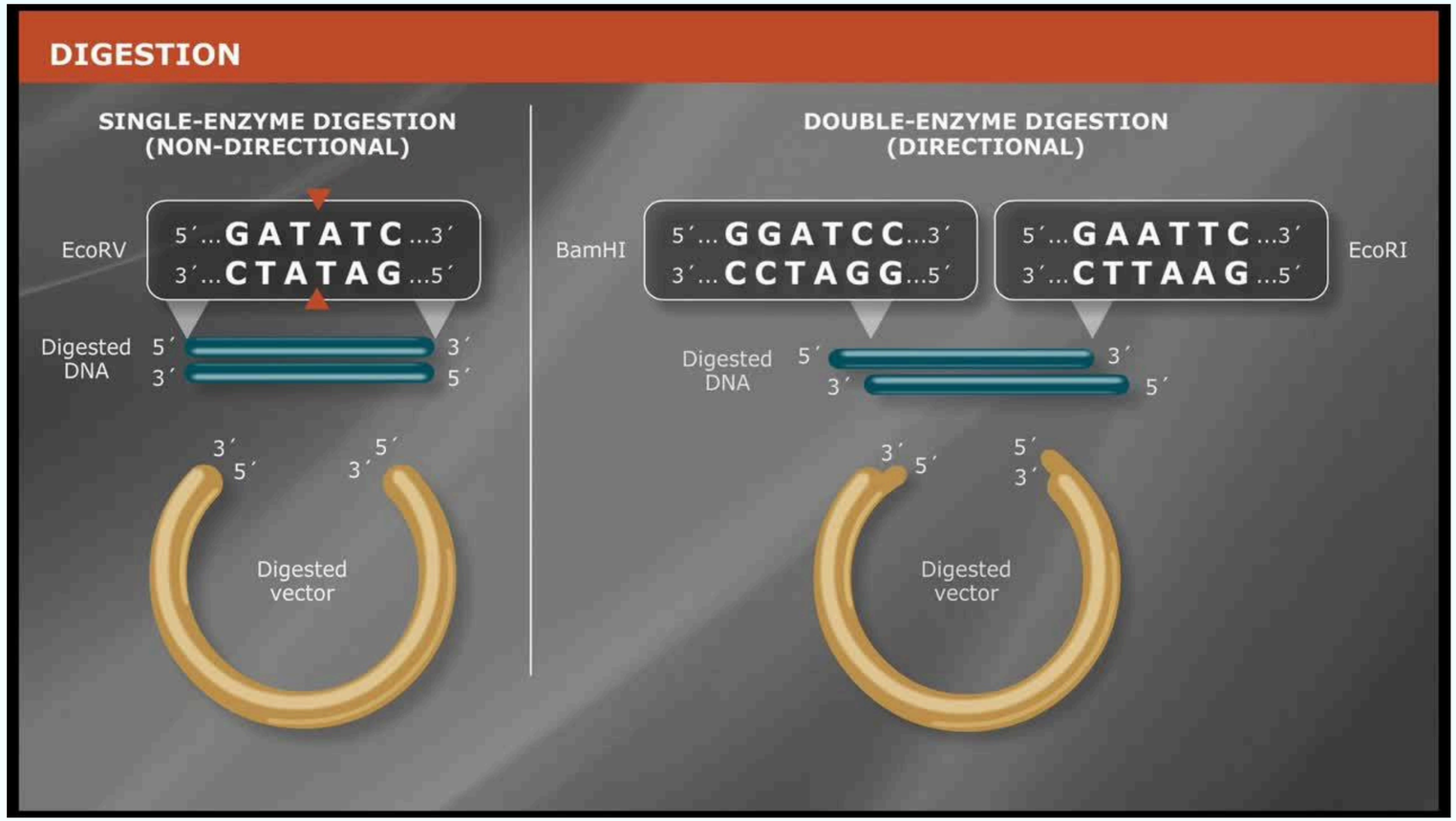
- The enzyme should have a unique restriction site in the vector's multiple cloning site (MCS).
- Ideally, it should not cut elsewhere in the vector.
- The restriction site should be positioned appropriately for insertion without disrupting essential vector elements.



Criteria for selecting restriction enzymes

Use two different restriction enzymes (directional cloning)

- Whenever possible, select two different restriction enzymes to create two distinct sticky ends



Advantages:

- Helps control insert orientation.
- Reduces vector self-ligation.
- Improves the efficiency of obtaining the desired construct.

Compatible DNA ends

Sticky ends, often facilitate efficient ligation

Blunt ends, can be ligated but generally less efficient

Compatible cohesive ends, different enzymes may generate compatible overhangs

Enzyme reaction condition

- Optimal reaction temperature.
- Buffer compatibility.
- Enzyme activity in the same buffer.
- Digestion time.
- Methylation sensitivity.

https://www.neb.com/en/products/restriction-endonucleases?srsltid=AYCX9vIe2-lMQ0_3PIHVho4GzLVfCYSGE9o2csixW_aPDcf8EcvWf5m2

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Home > Restriction Endonucleases > BamHI-HF®

BamHI-HF®

BamHI-HF has been reformulated with Recombinant Albumin (rAlbumin) beginning with Lot #10133983. Learn more.

5′... G▼GATCC...3′
3′... CCTAG▲G...5′

Isoschizomers | Single Letter Code | Pronunciation:

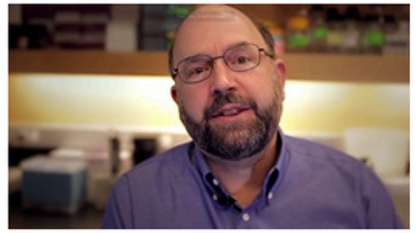
Not sensitive to dam, dcm or mammalian CpG methylation.

High-Fidelity (HF®) restriction enzymes have the same specificity as native enzymes, but have been engineered for significantly reduced star activity and performance in a single buffer (rCutSmart™ Buffer). All HF-restriction enzymes come with Gel Loading Dye, Purple (6X). Enjoy the enhanced performance and added value of our engineered enzymes at the same price as the native enzyme:

- Engineered for improved performance
- 100% activity in rCutSmart Buffer
- Time-Saver™ qualified for digestion in 5-15 minutes
- Reduced star activity
- Supplied with 1 vial of Gel Loading Dye, Purple (6X)
- Restriction Enzyme Cut Site: G/GATCC

Reagents Supplied | **Protocols**

Featured Video



Reduce Star Activity with High-Fidelity Restriction Enzymes

View additional videos

CRISPR’s Cas9

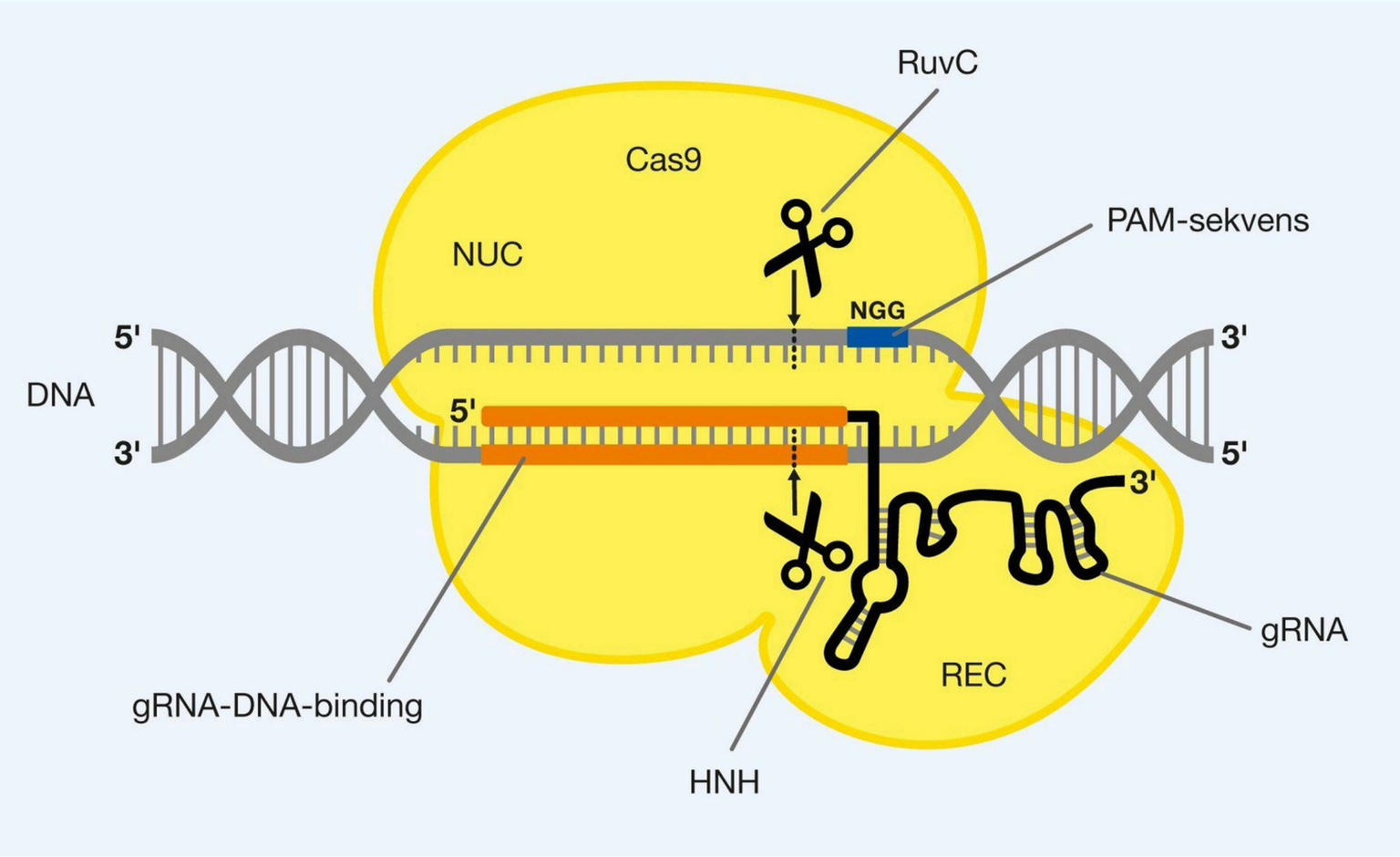
CRISPR - Clustered Regularly Interspaced Short Palindromic Repeats

Cas9 is an RNA-guided DNA endonuclease,

- a ~20nt guide sequence picks the target,
- PAM next to it licenses binding,
- two nuclease domains make a blunt double-strand break

CRISPR’s Cas9 Compoents

- Cas9, endonuclease that cuts DNA
- crDNA, contains the sequence that recognizes the target DNA
- tracrRNA, helps crRNA associate with Cas9 and supports complex formation

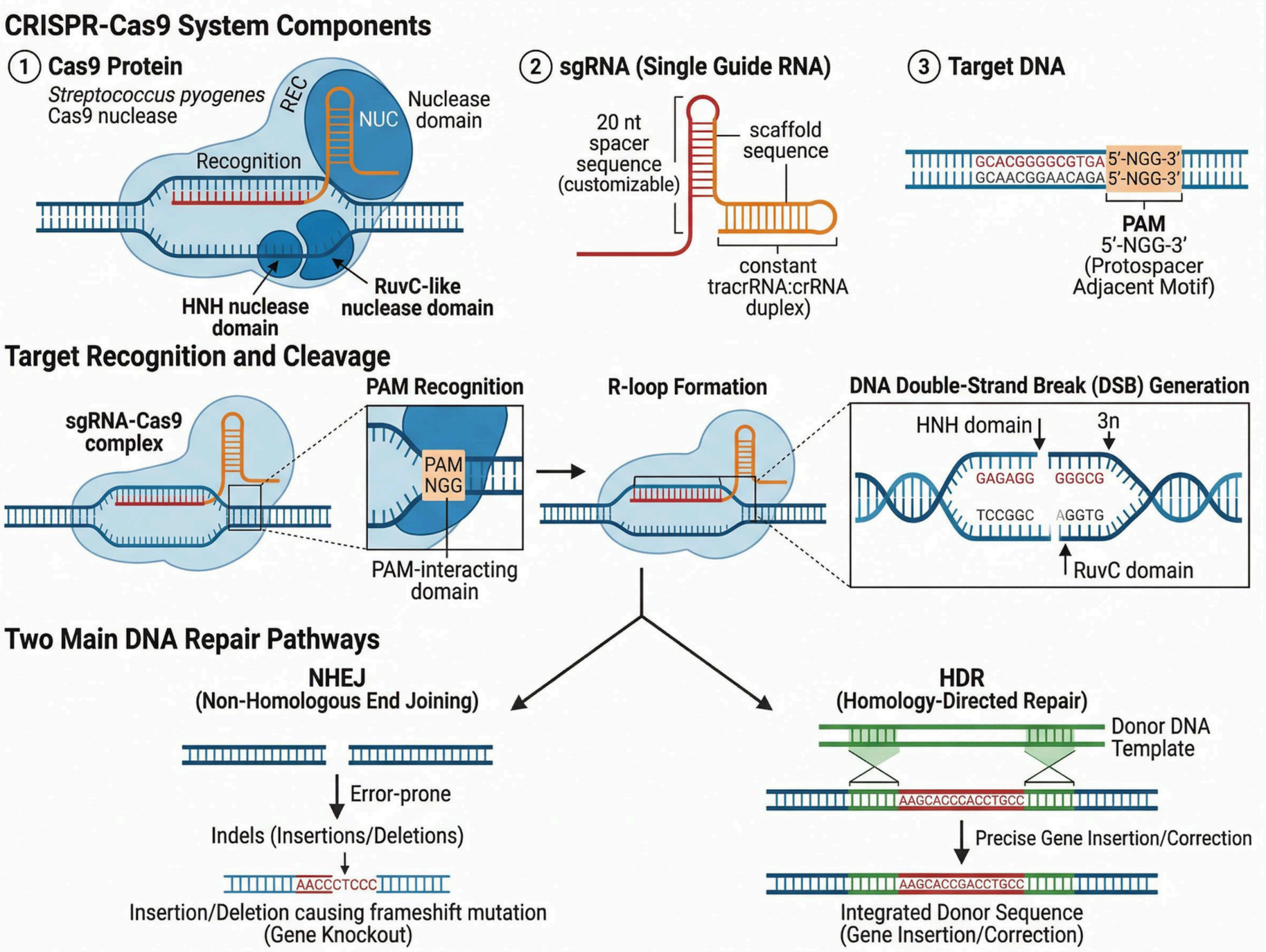


Non-homologuos and joining (NHEJ)

- Repairs DNA breaks without requiring a homologous repair template.
- Can introduce small insertions or deletions (indels).
- May disrupt the coding sequence and produce a gene knockout.

Homology-directed repair (HDR)

- Use a homologous DNA template.
- Can introduce a desired sequence modification.
- Can be used for precise sequence replacement or insertion (knock-in).



https://www.youtube.com/watch?v=4knIPgD6h1U

CRISPR-Cas9 versus traditional restriction enzymes

RE

- Recognize specific DNA motifs
- Determined by enzyme recognition sequence
- DNA cleavage
- Generates sticky or blunt ends
- App in molecular cloning

CRISPR-cas9

- Guide RNA-directed targeting plus PAM
- Can be programmed by changing sgRNA
- Genome editing
- Double-strand break
- Application: Genome engineering

Southern Blot

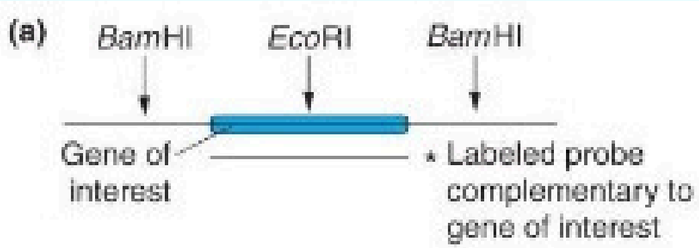
- Used to determine gene copy number;
- gene mapping;
- gene mutation detection;
- PCR product confirmation;
- DNA fingerprinting

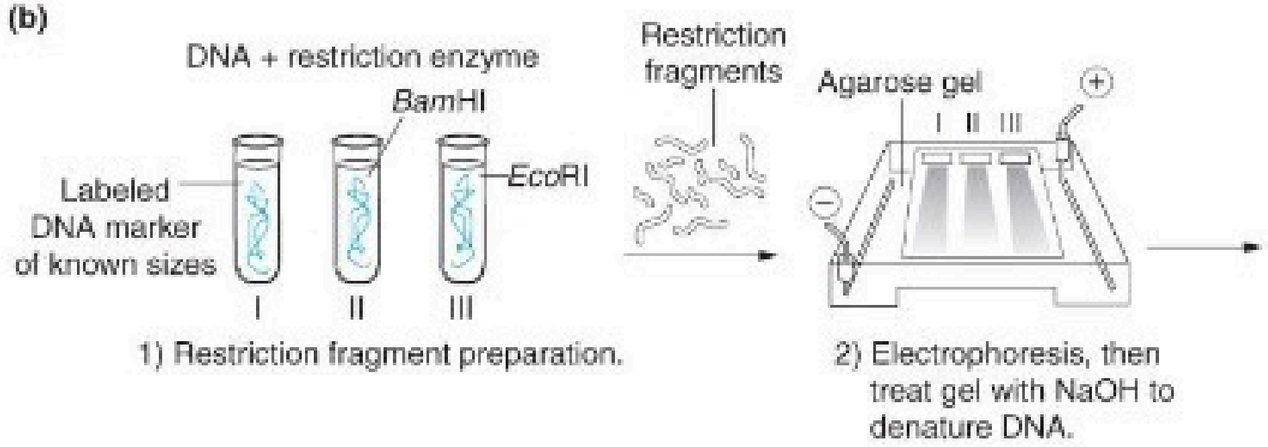
Principle:

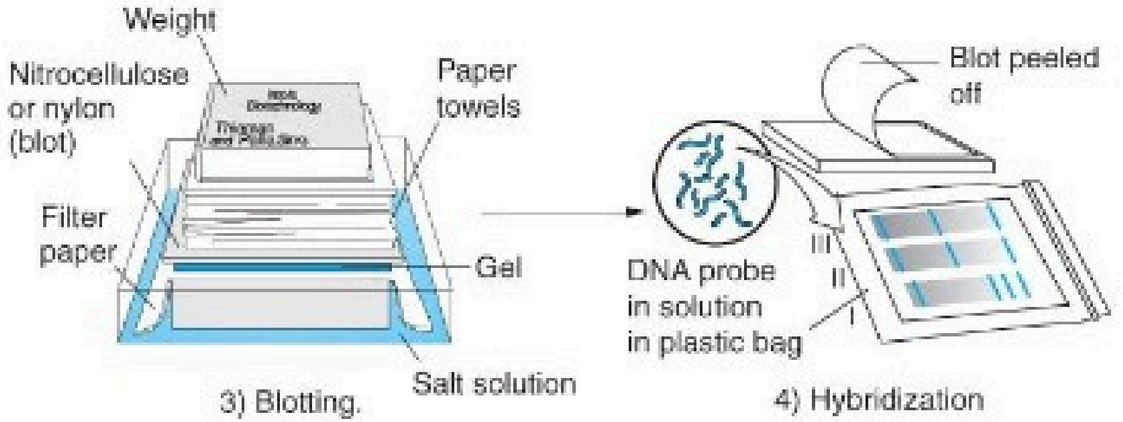
DNA–DNA hybridisation. Denatured single strands on the membrane re-pair with a complementary labelled probe; washing removes everything that did not pair well.

Steps

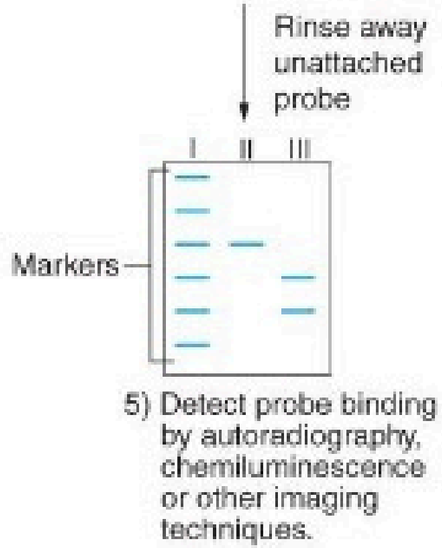
- Digest chromosomal DNA into small fragments with restriction enzymes
- Fragments are separated by agarose gel electrophoresis
- Gel is treated with alkaline solution to denature the DNA
- Fragments are transferred onto a nylon or nitrocellulose filter (called blotting)
- Filter (blot) is baked or exposed to UV light to permanently attach the DNA
- Filter (blot) is incubated with a labeled probe and exposed to film by autoradiography







https://www.youtube.com/watch?v=vBATs6Jhk44



Studying Gene Expression

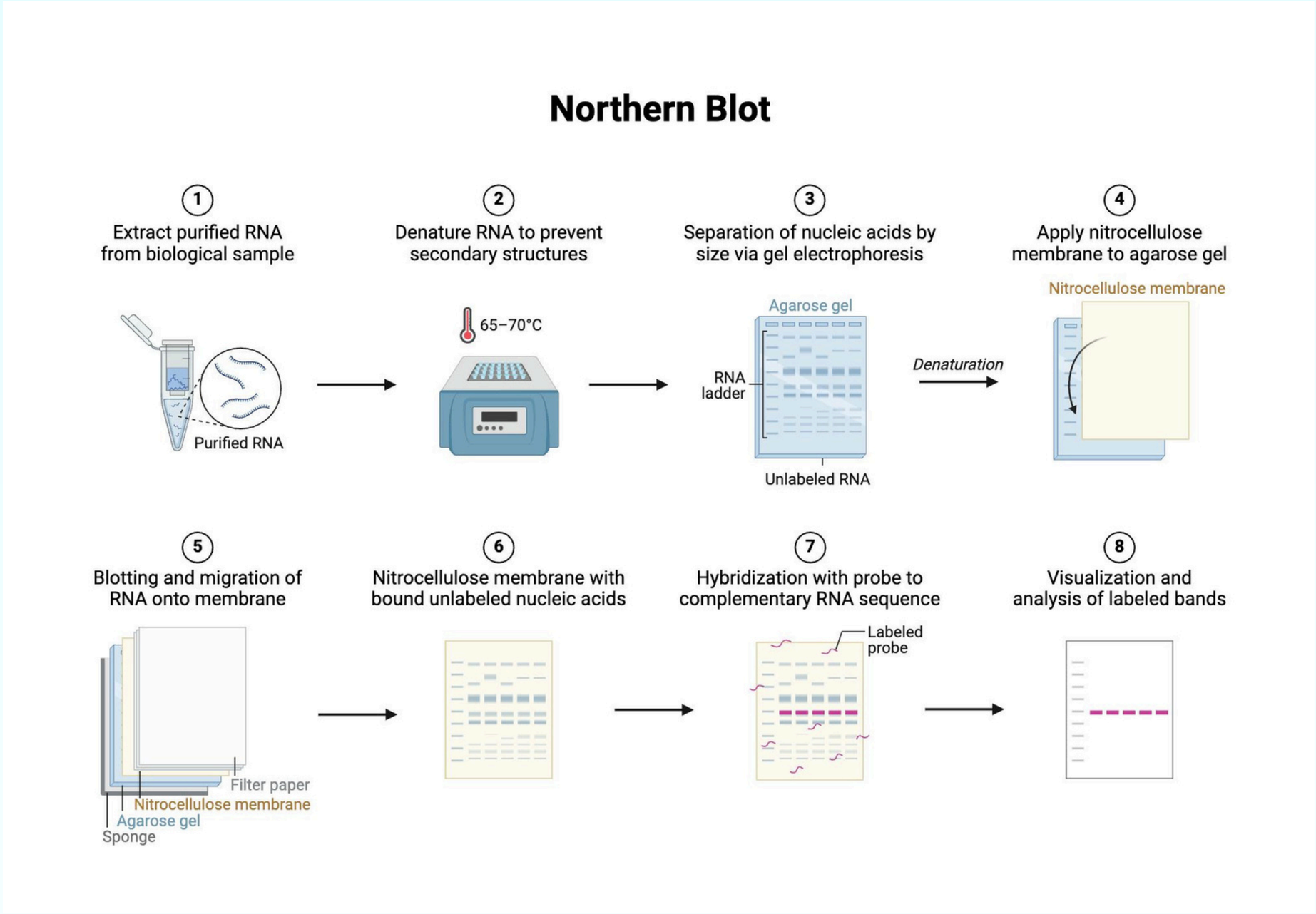
- Techniques involve analyzing mRNA produced by a tissue

Northern Blot Analysis

- Basic method is similar to Southern blotting
- To determine
  - Whether a particular RNA is present.
  - The approximate size of the RNA transcript.
  - The relative abundance of that RNA.
  - Differences in gene expression between samples or conditions.
- Can compare and quantify amounts of mRNA present in different tissues

Principle

- Separate RNA molecules according to their size.
- Transfer RNA from a gel onto a membrane.
- Use a labeled complementary DNA or RNA probe to detect the RNA of interest.
- Visualize the signal to determine whether the target RNA is present.



https://www.youtube.com/watch?v=G2Z1wuEOZUU

Steps

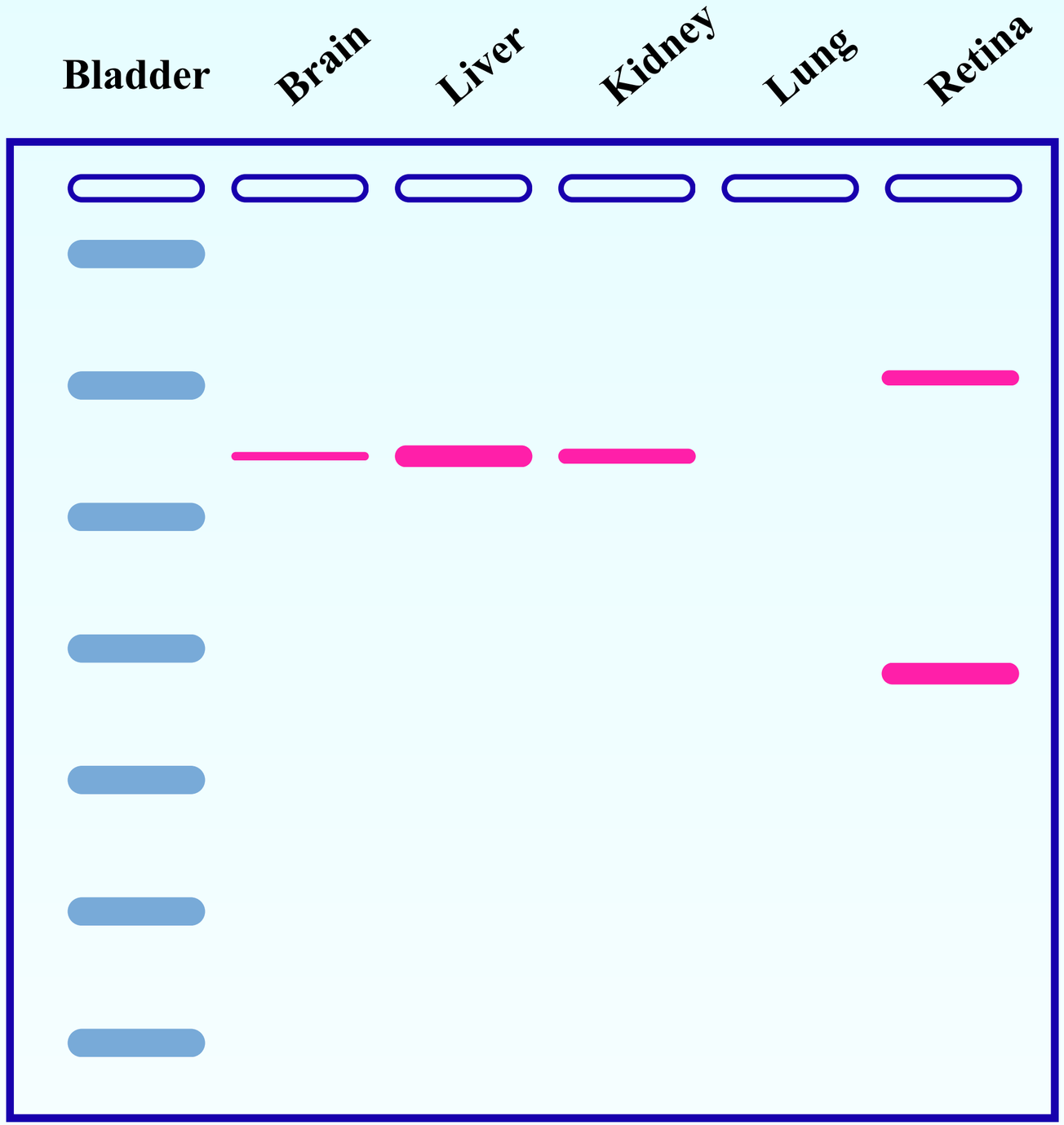
- RNA is isolated from a tissue of interest,
- separated by gel electrophoresis,
- blotted onto a membrane,
- hybridized to a labeled DNA probe,
- washing,
- signal detection,
- exposed bands on autoradiograph show presence of mRNA for gene of interest as well as size of mRNA

Chemicals different to Southern Blot

- No NaOH
- Denaturing gel instead
- No restriction digest
- RNase control

Result

- Band position => transcript size
- Band intensity => RNA abundance
  - Strong band: more target RNA
  - Weak band: less target RNA
  - No band: undetected target RNA
- Band pattern
  - Several band: alternative isoforms (splicing, polyadenylation)
  - Smear: degraded RNA
- Loading control => normalization (GAPDH or 28S/18S rRNA)
  - Confirms equal loading and transfer in each lane
  - Relative expression = target band / control band


Bladder
Brain
Liver
Kidney
Lung
Retina